

$$\begin{pmatrix} 1 & 4 & 5 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 52 \end{pmatrix}$$

if $n = 3$

$$O(n^3)$$

$$O(n^3)$$

$$\begin{array}{ccccccc} 1 & \times & 1 & + & 4 & \times & 4 & + & 5 & \times & 7 & = & 52 \\ & & & & & & 16 & & & & 35 & & \end{array}$$

$$\begin{pmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{00} & b_{01} & b_{02} \\ b_{10} & b_{11} & b_{12} \\ b_{20} & b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} c_{00} & c_{01} & c_{02} \\ c_{10} & c_{11} & c_{12} \\ c_{20} & c_{21} & c_{22} \end{pmatrix}$$

A
 B
 C

$$a_{00}b_{00} + a_{01}b_{10} + a_{02}b_{20} = c_{00}$$

$$a_{i0}b_{0j} + a_{i1}b_{1j} + a_{i2}b_{2j} = c_{ij}$$

$$c_{ij} = \sum_{k=0}^2 a_{ik}b_{kj} \quad \text{for all } i, j$$

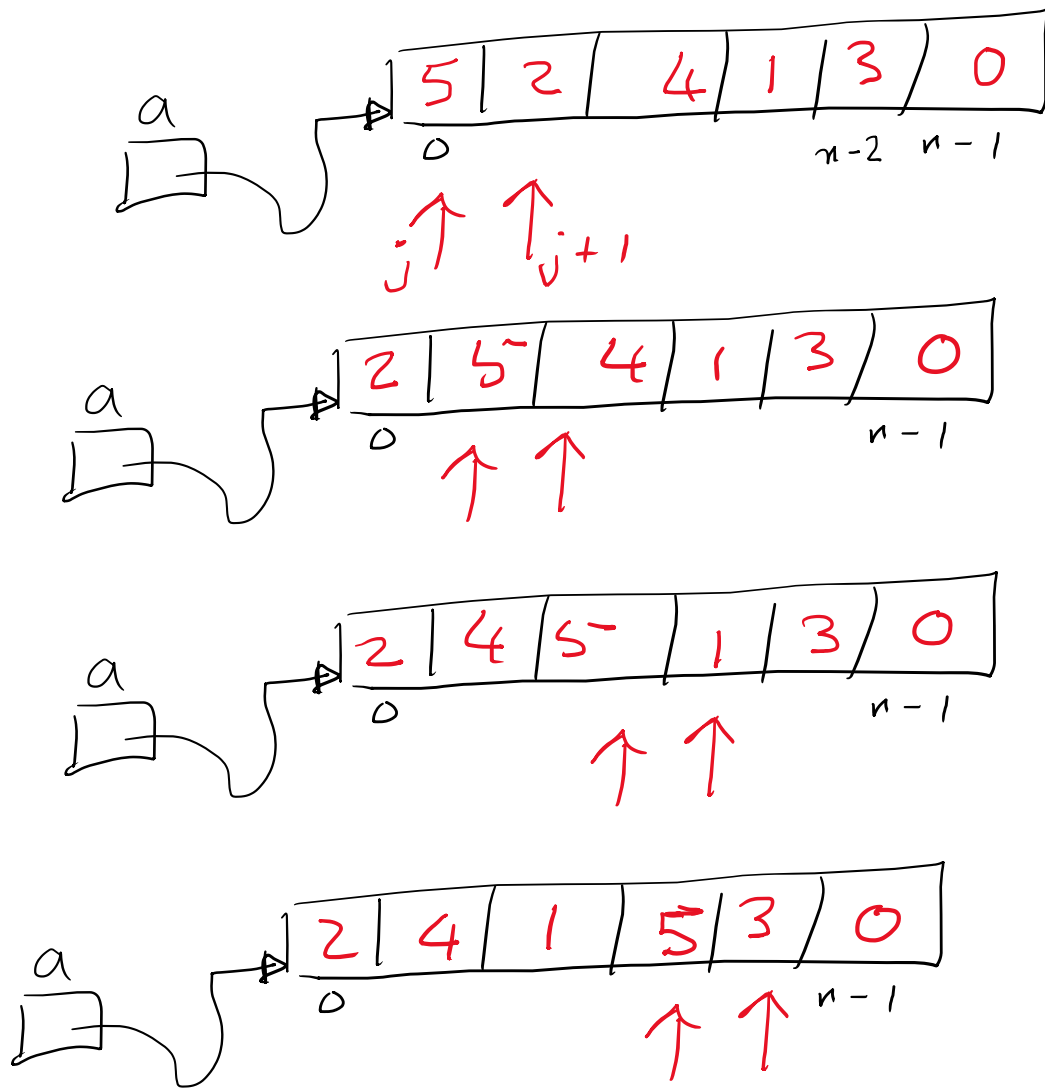
$$c_{ij} = \sum_{k=0}^2 a_{ik} b_{kj} \text{ for } \text{all } i, j \quad O(n^3)$$

```

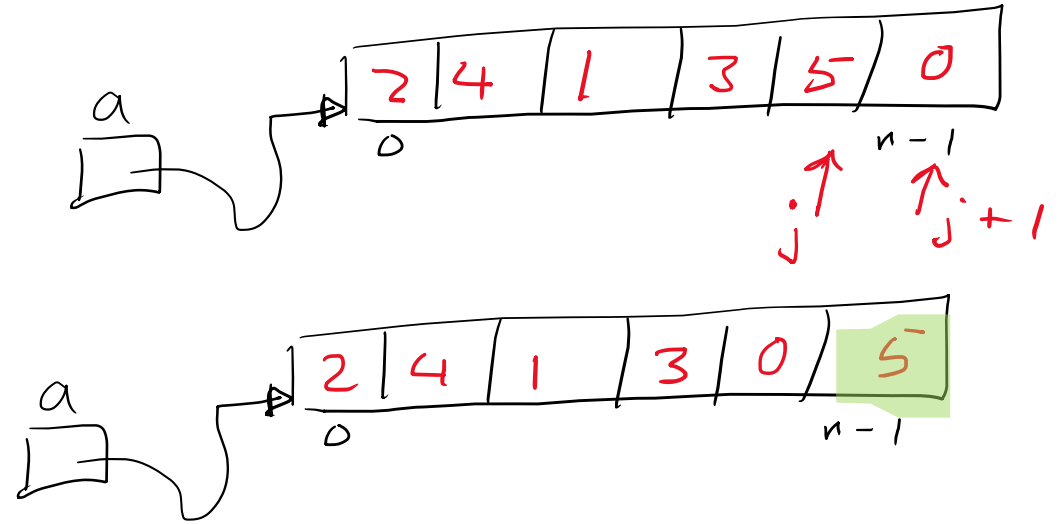
for (int i = 0; i < n; i++) { (n)
    for (int j = 0; j < n; j++) { (n)
        sum = 0.0;
        for (int k = 0; k < n; k++) { (n)
            sum = sum + a[i][k] * b[k][j];
        }
        c[i][j] = sum;
    }
}

```

expensive
cheap



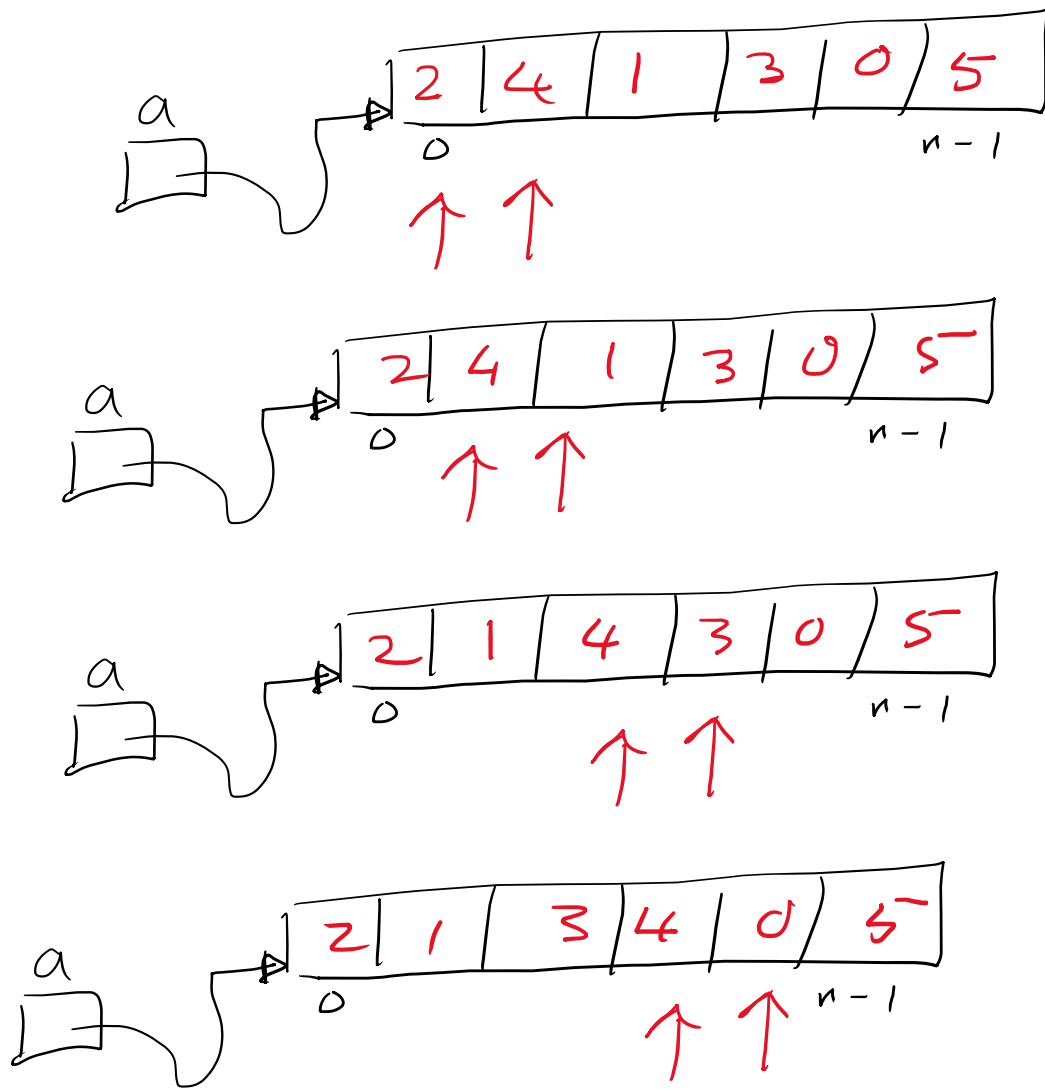
n items



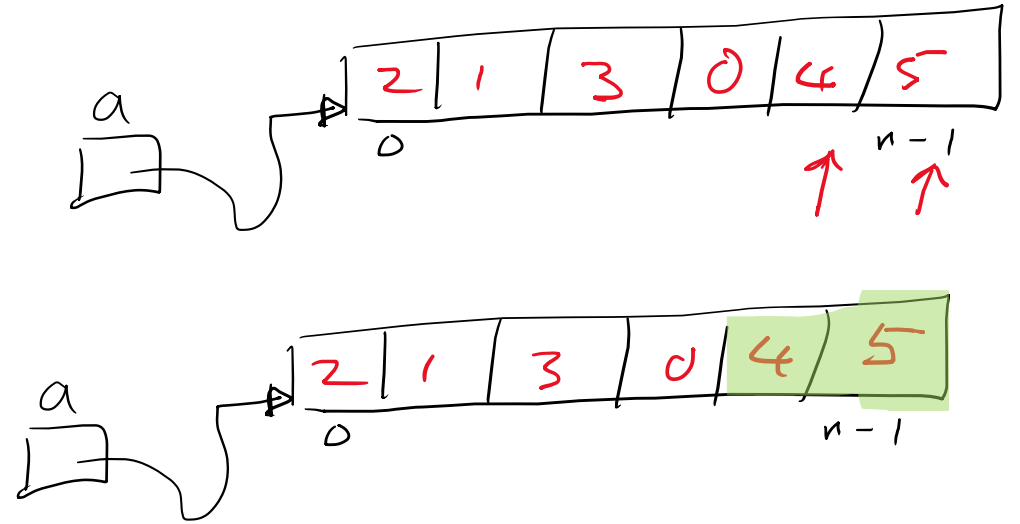
1st pass

$i = 0$

j starts at 0, and ends at $n-2$

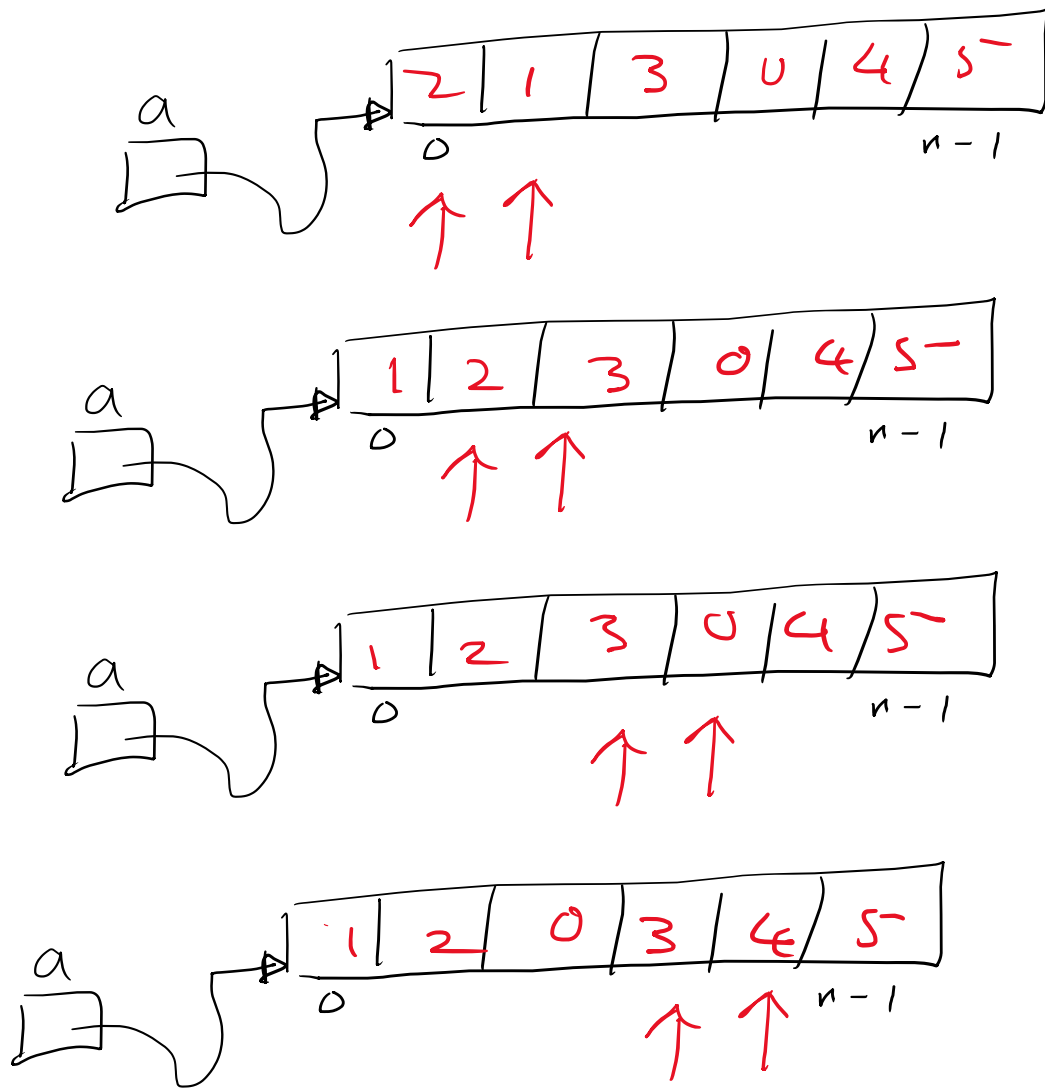


n items

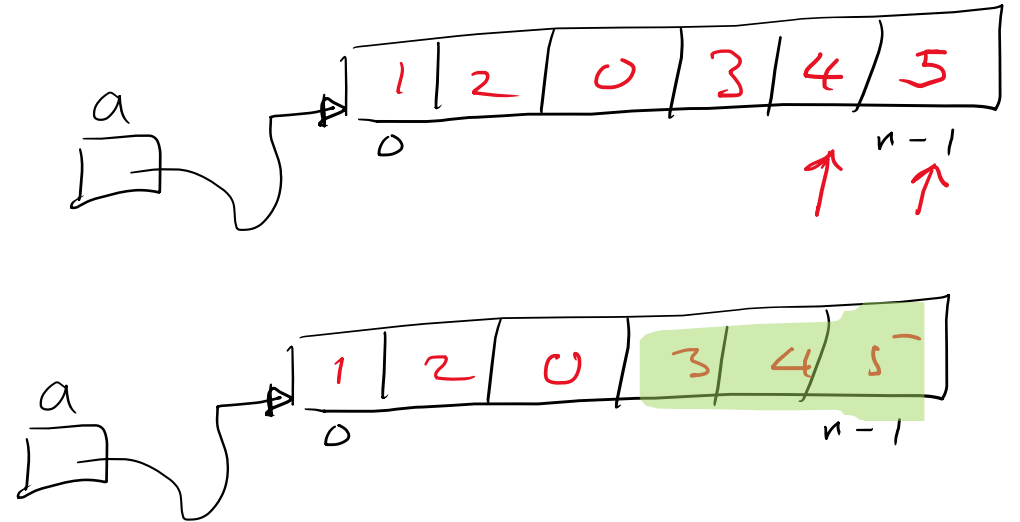


2nd Pass

$i = 1$



n items



3rd Pass

$$i = 2$$

Finish, you might be examined!

// Bubble Sort

for (int i = 0; $i < n-1$; i++) { $n-1$

for (j = 0; $j < n-1$; j++) { $n-1$

if ($a[j] > a[j+1]$)

temp = a[j]

a[j] = a[j+1];

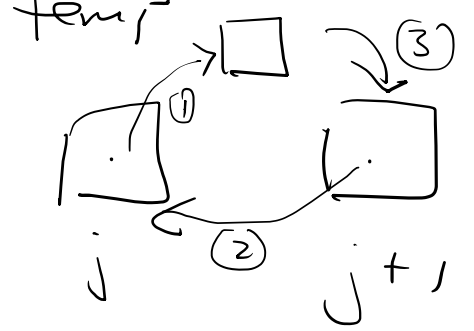
a[j+1] = temp;

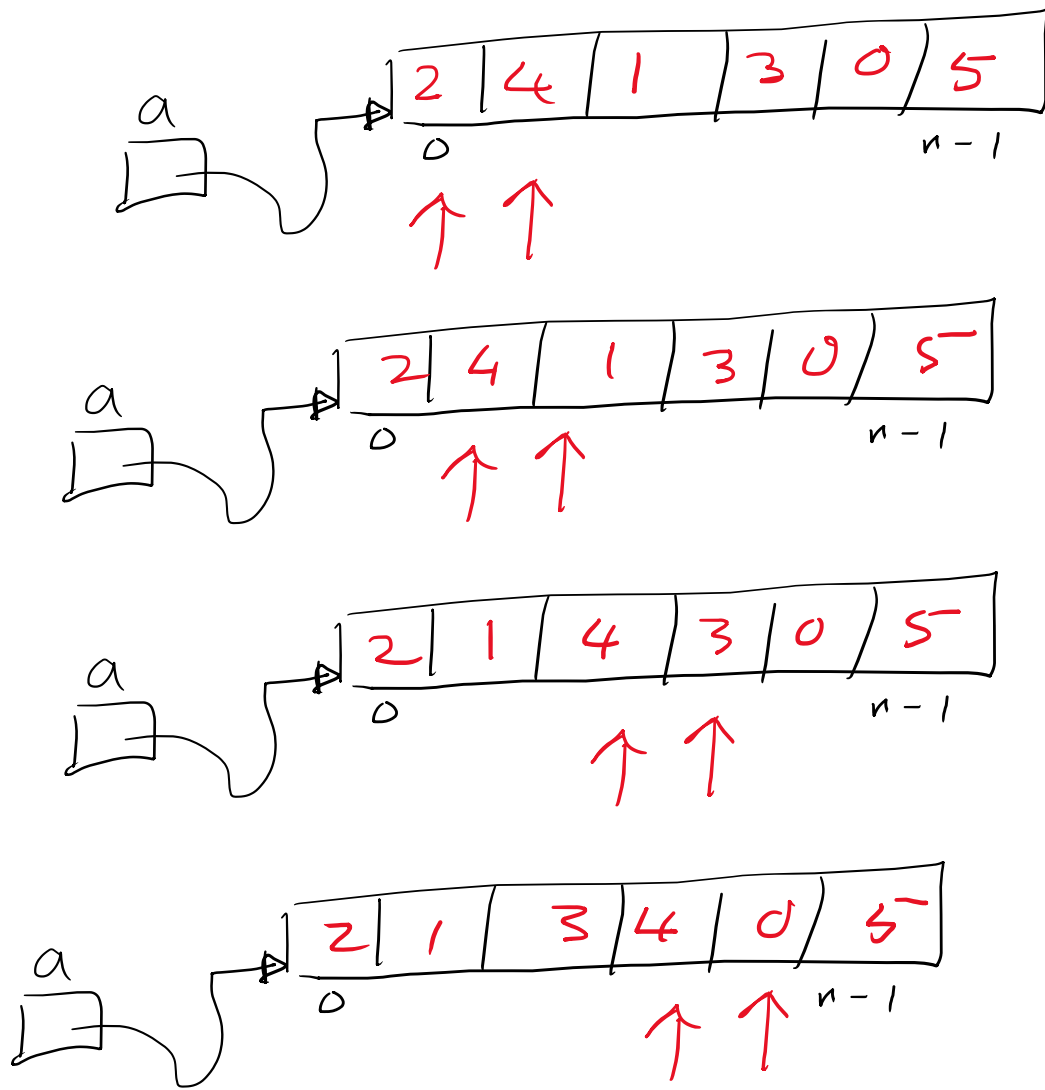
}

}

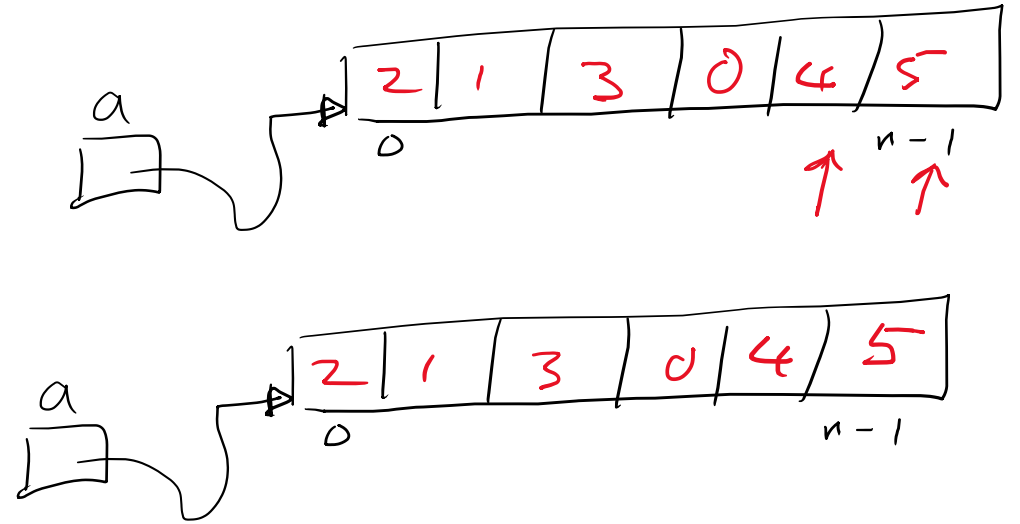
$$T(n) = (n-1)(n-1) = n^2 - 2n + 1$$

$O(n^2)$





n items



2nd Pass

- So, given that you want to sort 1000 numbers you would need to perform in the region of 1 000 000 comparisons. This could be very time consuming.

- $O(n \log n)$ – QuickSort

$$\log_{10} 1000 = 3, \quad \text{because } 10^3 = 1000 = 10 \times 10 \times 10$$

$$T(n) = n \log n = 1000 \times 3 = 3000$$

